

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

Section 5

Mains cables

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Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

1. GENERAL

All system-parts, except the multi image camera, the transformer for the compressor, the optional chiller for the refrigerator system are and the optional Easy Vision are powered via mains distribution unit.

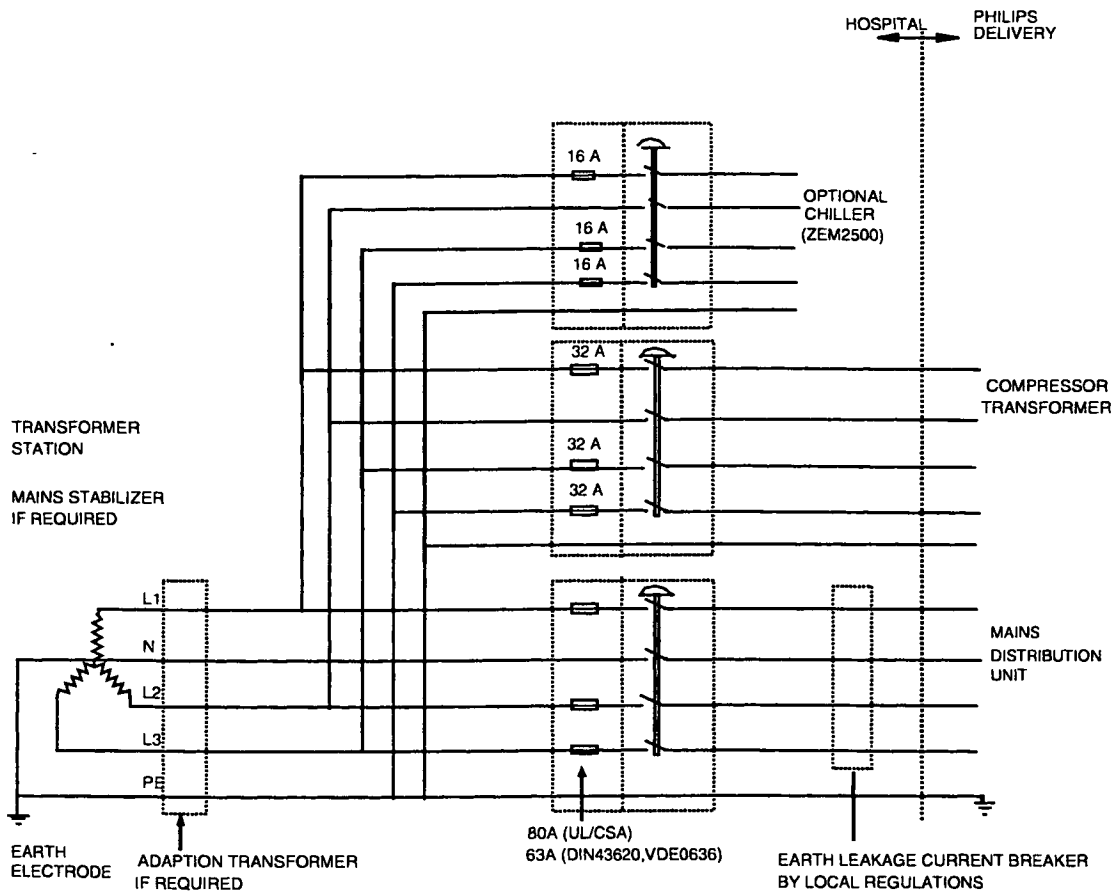
The power line must be provided with a mains switch and fuses/circuit-breakers, installed near to the M.D.U.

The cables should be routed in accordance to the Site planning drawings.

All cables are delivered with the system except for the mains cable to the mains distribution cabinet.

The transformer of the compressor receives its supply from the same source as the mains distribution unit.

Figure 1 - Mains connections



FH92 DRW

For USA only

The refrigerator system transformer is not delivered with the system and is not required.

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

2. CABLE DUCTS

The ducts inside the technical room must have three separate compartments:

- | | | |
|---|--|--------------|
| 1 | Mains and earth cables | (50x100 mm) |
| 2 | Gradient cables and magnet ramp cables | (150x100 mm) |
| 3 | Signal cables | (200x100 mm) |

For the ducts inside the technical room the minimum dimensions are 80x400 mm.

The ducts in the examination room must have a minimum length of 4 meters so that extra cable length can be stored. The ducts must be located between the rear of the magnet and the system filter box.

3. CONNECTING CABLES TO THE MDU

CAUTION

Check the mains voltage of the incoming and outgoing connectors of the Mains Distribution Unit. Measure with a multimeter the incoming line HX0 and the outgoing lines HX1..X10 (must be 220 Volt). After checking this, switch OFF the power in the MDU by switching OFF all switches/fuses including S1.

Connect all mains cables according (Z5-2.8).
Connect all earth cables according (Z5-2.9).

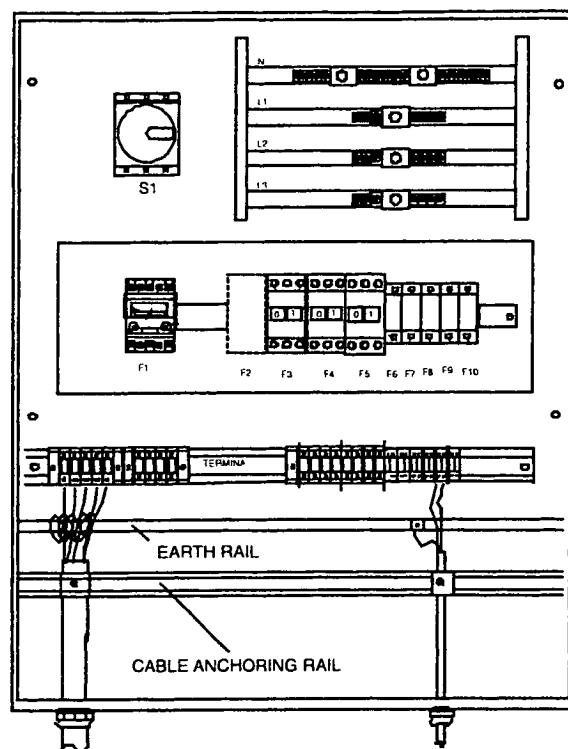
All cables to the MDU must be stripped by removing 10 cm of the outer insulation and 1 cm of the insulation around the wire. Pass the cables via the openings in the bottom of the MDU. Mount the relief brackets.

Cable color code:

Blue	=	Neutral
Black/brown	=	Phase
Green/yellow	=	Earth/ground

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

Figure 2 - Mains Distribution Unit

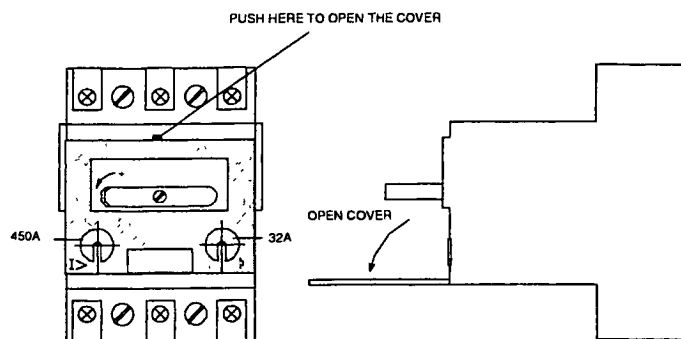


FH/81 WPC

Check/adjust

F3	Thermal current setting	25 A
F4	Thermal current setting	25A
F5	Thermal current setting	16A
F1	Thermal current setting	32 A.
F1	Magnetic current setting	450 A.

Figure 3 - Adjustment of F1



Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

Mains connections with MPS ramp tool

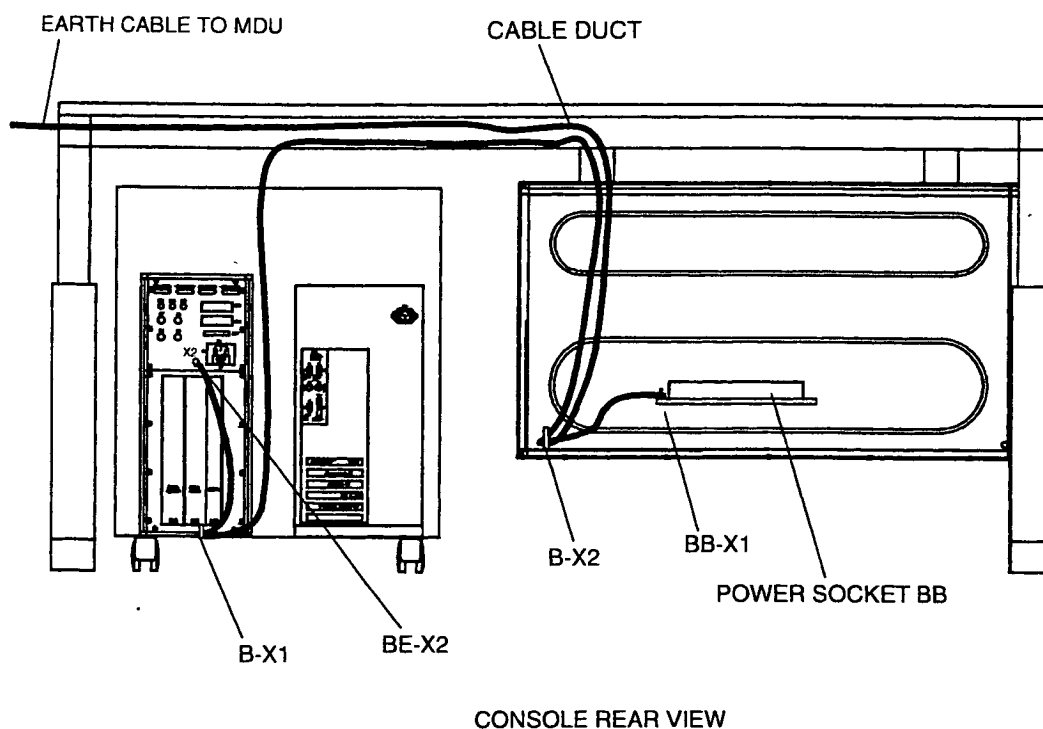
SWITCH FUSE	USED FOR	FUSE (A)	TERMINAL
S1	F2...F12 (System)	63	HX0
F1	Gradient amplifier	40	HX1
F2	(empty slot)		
F3	NTDAC	25	HX3
F4	Magnet Power Supply ramp tool	25	HX4
F5	Console	16	HX5
F6	Front End	16	HX6
F7	Spare	16	HX7
F8	Spare	16	HX8
F9	Spare	16	HX9
F10	RF amplifier S23/S24	32	HX10

Table 1: MDU switches and terminals with MPS ramp tool

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

4. EARTH CABLES AT THE CONSOLE

Figure 4 – Earth cables at the console



5. MPS RAMP TOOL

With the initial system, a 15-meter long mains cable with a 3-phase connector is supplied that must be used to connect the MPS ramp tool to the MDU. After installation this mains cable must be stored on the site.

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

6. CONNECTING THE REFRIGERATOR COMPRESSOR

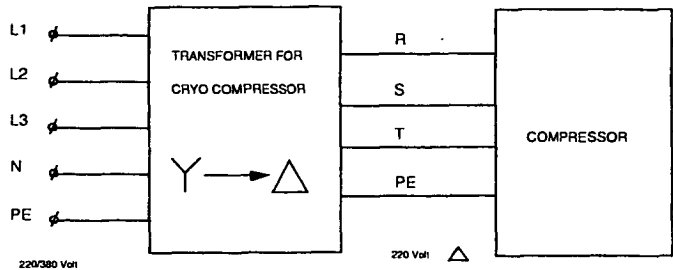
The cryo-cooler compressor is supplied by a dedicated 3 phase 380V or 400V power line.

The line shall be provided with a mains switch and fuses (3 x 32 Slow Blow).

Connections of the transformer:

- In-coming mains L1, L2, L3, N, earth.
- Out-going mains to compressor L1, L2, L3, earth.

Figure 5 - refrigeratorcompressor mains connection



The plastic lips in transformer cabinet must be removed in order to lock the door with the delivered keys.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

Section 5

Mains cables

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Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

1. GENERAL

For the Gyroscan Intera Power and Master a Mains Distribution Extension (HA) will be delivered. The hospital mains will be connected to this unit. At the output, the Mains Distribution Unit (H) will be connected. Also the Copley 274 will be connected to the MDU extension. For the Gyroscan Intera Master also the second copley 274 will be connected to this unit. The Hard Copy Unit (HCU), the transformer of the compressor, the Heat exchanger and the chiller for the refrigerator system will **not** be powered via Mains Distribution Extension or Mains Distribution Unit.

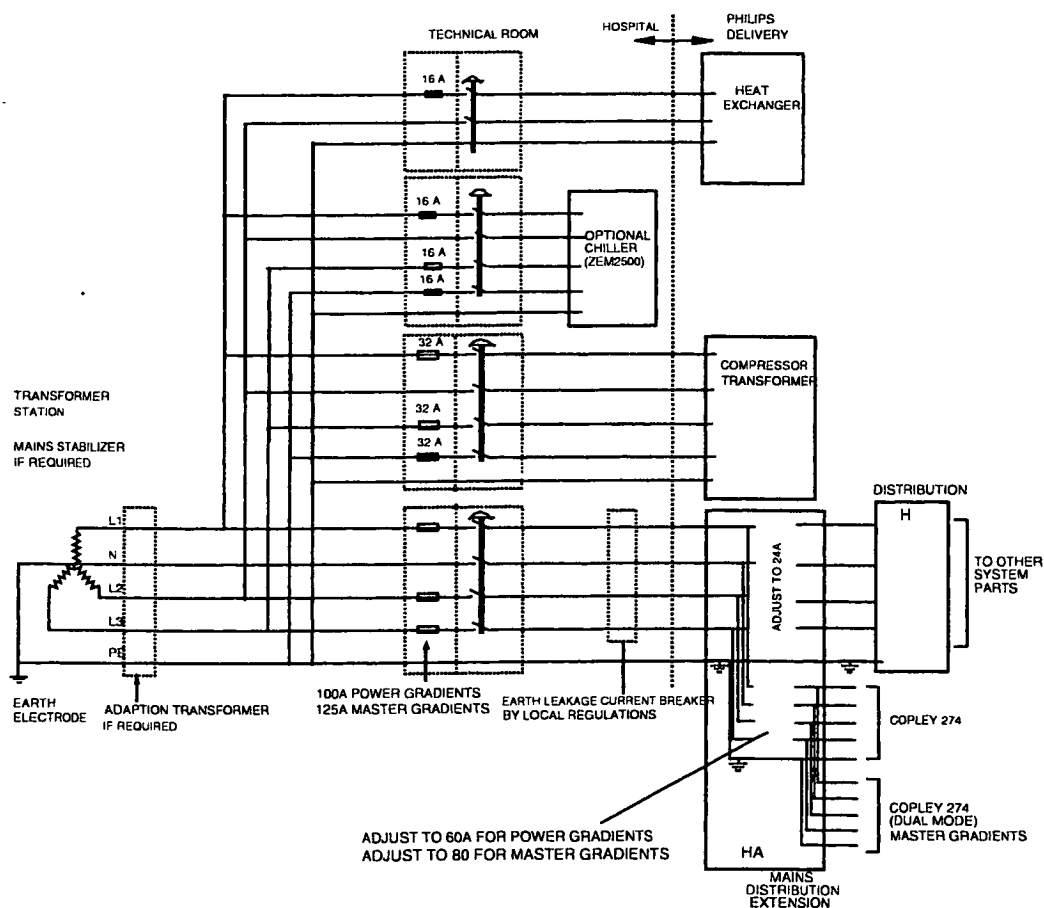
The power line must be provided with a mains switch and fuses/circuit-breakers, installed near to the MDU extension.

The cables should be routed in accordance to the Site planning drawings.

All cables are delivered with the system except for the mains cable to the MDU extension.

The transformer of the compressor receives its supply from the same source as the mains distribution unit.

Figure 1 - Mains connections



FH93 DRW

For USA only

The refrigerator system transformer is not delivered with the system and is not required.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

2. CABLE DUCTS

The ducts inside the technical room must have three separate compartments:

- | | | |
|---|--|--------------|
| 1 | Mains and earth cables | (50x100 mm) |
| 2 | Gradient cables and magnet ramp cables | (150x100 mm) |
| 3 | Signal cables | (200x100 mm) |

For the ducts inside the technical room the minimum dimensions are 80x400 mm.

The ducts inside the examination room must have two separate compartments:

- | | | |
|---|------------------|-------------|
| 1 | Gradient cables | (80x200 mm) |
| 2 | All other cables | (80x400 mm) |

The ducts in the examination room must have a minimum length of 4 meters so that extra cable length can be stored. The ducts must be located between the rear of the magnet and the system filter box.

3. CONNECTING CABLES TO THE MDU'S

CAUTION

Check the mains voltage of the incoming and outgoing connectors of the Mains Distribution Unit. Measure with a multimeter the incoming line HX0 and the outgoing lines HX1...X10 (must be 220 Volt). After checking this, switch OFF the power in the MDU by switching OFF all switches/fuses including S1.

For Gyroscan Master/Power connect all mains cables according (Z5-2.8).
For Gyroscan Master/Power connect all earth cables according (Z5-2.9).

All cables to the MDU must be stripped by removing 10 cm of the outer insulation and 1 cm of the insulation around the wire. Pass the cables via the openings in the bottom of the MDU. Mount the relief brackets.

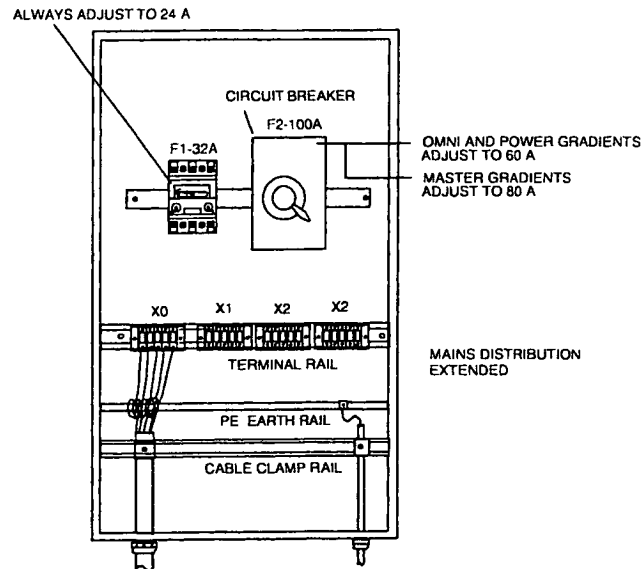
Cable color code:

Blue	=	Neutral
Black/brown	=	Phase
Green/yellow	=	Earth/ground

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

4. MAINS DISTRIBUTION UNIT EXTENSION (HA)

Figure 2 - MDU extension (HA)

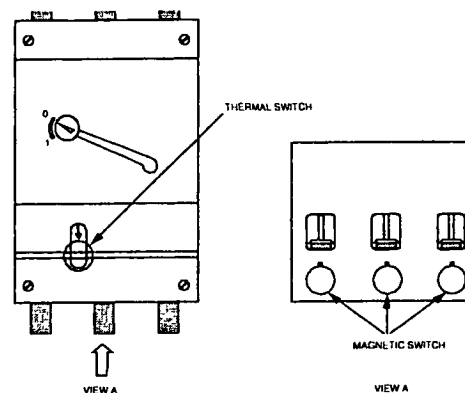


FH130 DRW

Check/adjust

F1	Thermal current setting	24 A
F1	Magnetic current setting	325 A
F2	Thermal current setting	60 A (For Gyroscan Intera Power)
F2	Thermal current setting	80 A (For Gyroscan Intera Master)
F2	Magnetic current setting	600 A (For Gyroscan Intera Power and Master)

Figure 3 - Adjustment of F2



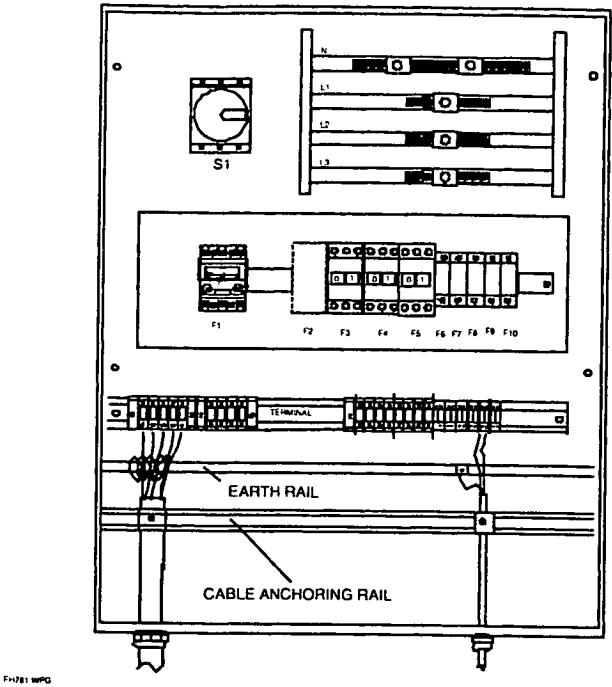
SWITCH FUSE	USED FOR	FUSE (max. A)	TERMINAL
F1	Incoming Hospital mains	32	HAX0
F2	To Mains Distribution Unit	100	HAX1
F2	Gradient amplifier (Gyroscan master (master))	100	HAX2
F2	Gradient amplifier (Gyroscan Power (Slave))	100	HAX2

Table 1: MDU extension switches and terminals

Intera 0 5 T	Intera 1 0 T	Intera 1 5 T
	Power	Power Master

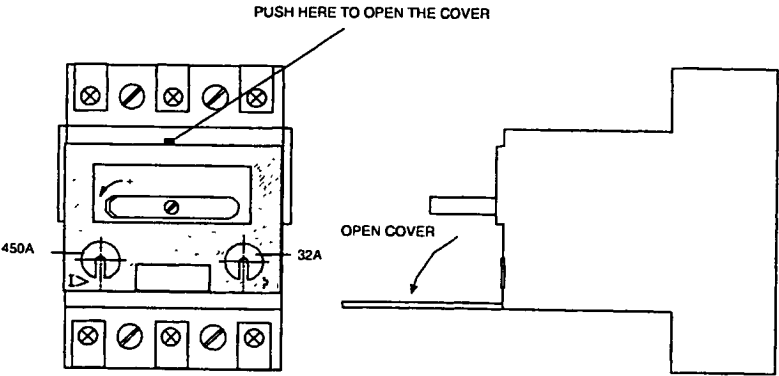
4.1. MAINS DISTRIBUTION UNIT (H)

Figure 4 - Mains Distribution Unit



Check/adjust		
F3	Thermal current setting	25 A
F4	Thermal current setting	25 A
F5	Thermal current setting	16 A
F1	Thermal current setting	32 A.
F1	Magnetic current setting	450 A

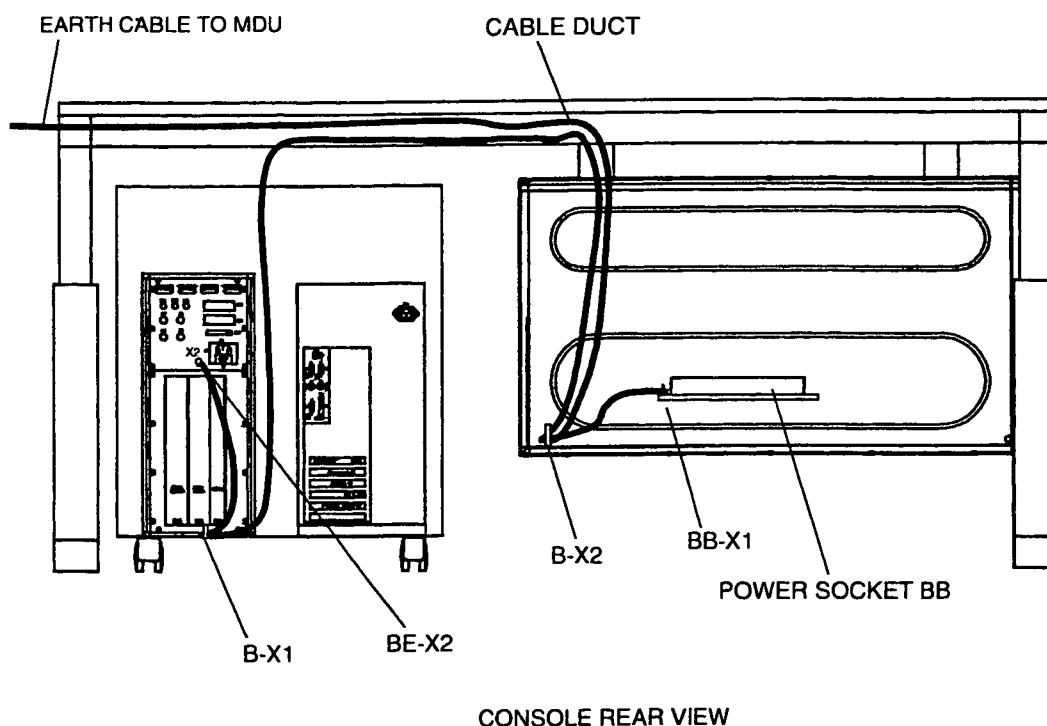
Figure 5 - Adjustment of F1



Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

Mains connections with MPS ramp tool

SWITCH FUSE	USED FOR	FUSE (A)	TERMINAL
S1	F2...F12 (System)	63	HX0
F1	(empty slot)	40	HX1
F2	(empty slot)		
F3	NTDAC	25	HX3
F4	Magnet Power Supply ramp tool	25	HX4
F5	Console	16	HX5
F6	Front End	16	HX6
F7	Spare	16	HX7
F8	Spare	16	HX8
F9	Spare	16	HX9
F10	RF amplifier S23/S24	32	HX10

Table 2: MDU switches and terminals with MPS ramp tool**5. EARTH CABLES AT THE CONSOLE****Figure 6 – Earth cables at the console**

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

6. MPS RAMP TOOL

With the initial system, a 15-meter long mains cable with a 3-phase connector is supplied that must be used to connect the MPS ramp tool to the MDU. After installation this mains cable must be stored on the site.

7. CONNECTING THE REFRIGERATOR COMPRESSOR

The refrigerator compressor is supplied by a dedicated 3 phase 380V or 400V power line.

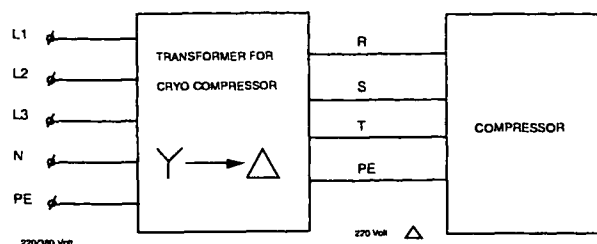
The line must be provided with a mains switch and fuses (3 x 32 Slow Blow).

Connections of the transformer:

In-coming mains L1, L2, L3, N, earth.

Out-going mains to compressor L1, L2, L3, earth.

Figure 7 - Mains connection to cryo-cooler compressor



The plastic lips in transformer cabinet must be removed in order to lock the door with the delivered keys.

8. 0V REFERENCE CABLE DUAL-MODE MASTER GRADIENTS ONLY

A white 50mm² 0V-reference cable must be installed between DX0 (master) to DX0 (slave) according the next figure. The terminals DX0 can be found in the rear bottom of both copley 274-gradient amplifiers. Also connect the white 0V-reference cable between DX0 (master) and ZX41 (System Filter Box).

Figure 8 - Gyroscan Intera Master master-slave earth connection

